

A Variational and Field-Theoretic Formulation of Neural Architecture Meta-Optimization

Least Complexity, Least Time, and Criticality

Analytical reference notes

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Abstract

We give a self-contained account of a family of meta-optimization formulations for neural networks, in which the choice of *width*, *depth*, and ultimately *architecture* is cast as a variational problem over measures and fields. The organizing principle is physical: the training objective is read as an *action*, its minimizers as *least-action trajectories*, and the associated statistical ensemble as a *path integral* over network configurations. We present the material from the general to the specific. We first state the master description-length principle and its causally ordered rung hierarchy—kinematics $\rightarrow$ degrees of freedom $\rightarrow$ dynamics $\rightarrow$ observables—the order in which a physicist writes down an effective theory (symmetry before dynamics, ultraviolet before infrared). We then descend the hierarchy one rung at a time. The outermost *kinematic* rung is the choice of computational contract (the contract and its symmetry); it is intrinsically discrete, because distinct contracts have incompatible interfaces and cannot be relaxed into one another, and it is settled by a description-length bake-off. The *degrees-of-freedom* rungs then fix the size of the network in that contract: **Scenario 0** (fixed architecture) lifts two-layer training to a convex problem over a measure of neurons, whose optimality condition is a dual certificate guaranteeing a finite-width (atomic) solution, and whose dynamics are a Wasserstein gradient flow of a free energy; **Scenario 1** promotes *depth* to an optimized quantity, in two Legendre-dual ways: *Route 2*, a grand-canonical ensemble that prices layers with a chemical potential μ and yields an optimal depth $L^* \sim \mu^{-1/(\alpha+1)}$; and *Route 1*, a reparameterization-invariant continuum-depth (neural-ODE) formulation in which complexity becomes arc length in a learned metric and the least-complexity network is a geodesic. A renormalization-group (RG) analysis [5] of the depth map supplies a criticality condition and a tower of critical exponents that make these depth objectives computable. The innermost *dynamics* rung is the trained weights as a saddle of the action, with finite width appearing as loop corrections; the *observable* rung reads off the derived architecture parameter. Throughout we make explicit which structures are exact (two-layer convexity, the dual certificate, finite atomicity) and which degrade with depth (joint convexity, the fixed-operator certificate), and we identify criticality [6, 7] as the locus where the depth objectives become well posed. We close with an honest ledger: the construction is a faithful *synthesis* of established results, and its sharpest depth-scaling predictions are heuristic and, where tested, not empirically confirmed.

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1 Introduction and physical viewpoint

The premise of these notes is that the minimum-description-length (MDL) view of learning — *find the simplest model consistent with the data*—is naturally expressed as a variational principle, and that the variational principle has the same formal structure as least-action mechanics and its quantum (path-integral) completion. Under this dictionary the depth index ℓ of a network plays the role of *time*, the composition of layers plays the role of *time ordering*, the trained network is a *classical trajectory* (a saddle point of an action), and finite-width fluctuations are *loop corrections* in a systematic expansion. The dictionary is summarized in Table 1.

These notes are organized from the general to the specific, in the order a physicist would impose on an effective theory rather than in the order the pieces were historically assembled. §2 states the master variational principle and its causally ordered rung hierarchy (kinematics $\rightarrow$ degrees of freedom $\rightarrow$ dynamics $\rightarrow$ observables) in full. The subsequent sections descend that hierarchy one rung at a time: §3 treats the outermost *kinematic* rung (which computational contract, i.e. which contract and symmetry, the datum lives in); the two *degrees-of-freedom* sections then fix the size of the network in that contract—§4 the *width* at fixed architecture (**Scenario 0**), and §6 the *depth* promoted to an optimized variable (**Scenario 1**), via a discrete priced route (Route 2, §6.1) and a

Learning object	Physical analogue
Regularized training objective	Action S / free energy $\mathcal{F}$
Trained network	Classical (least-action) trajectory
Depth index ℓ	Time t
Layer composition	Time ordering / transfer operator
Backpropagation	Adjoint (costate) equation, Pontryagin
Infinite depth	Continuum (path-integral) limit
Infinite width	Mean field / classical limit
Neural tangent kernel (NTK)	Gaussian (free) propagator
Finite-width correction	Loop expansion in $1/m$
Depth-wise signal propagation	Renormalization-group flow

Table 1: The physical dictionary underlying the formulations. Each learning object is placed in correspondence with a mechanical/field-theoretic one; the correspondences are exact where noted and heuristic otherwise.

continuum geometric route (Route 1, §6.2); §7 treats the *dynamics* (the trained weights as a saddle of an action, with finite width as loop corrections); and §9 reads off the *observables* (the derived architecture parameter). A *criticality* thread (§8) runs through the depth analysis and makes its objectives computable through a single critical exponent. The three formulations of increasing internal scope are thus:

- **Scenario 0** (§4): fixed architecture; the free variable is the *width* (number of neurons), realized as a measure over neuron parameters.
- **Scenario 1** (§6): *depth* is promoted to an optimized variable, via a discrete priced route (Route 2, §6.1) and a continuum geometric route (Route 1, §6.2).
- The *criticality* thread (§8), which threads through both and makes the depth objectives computable through a single critical exponent.

Notation is standard. Write a two-layer network with hidden width m as $f(x) = \frac{1}{m} \sum_{j=1}^m a_j \sigma(u_j^\top x)$, with neuron parameters $\Theta_j = (u_j, a_j) \in \Theta$ and activation σ . Here $\mathcal{M}(\Theta)$ denotes finite signed measures on Θ , and $\mathcal{P}(\Theta)$ probability measures; $\rho \in \mathcal{M}(\Theta)$ is the neuron measure whose mass plays the role of width (and $\rho^{(\ell)}$ its per-layer copy in a deep net). We write $R[\cdot]$ for a data-fitting risk and reserve λ, μ for the chemical potentials (prices) conjugate to width and depth respectively, and β for an inverse temperature. The subscript $\|\cdot\|_{\text{TV}}$ denotes the total-variation (atomic) norm of a measure, the sparsity penalty whose minimizer is atomic (finitely many neurons).

2 The variational principle and its rung hierarchy

We state the organizing principle in full at the outset, and devote the rest of the notes to deriving each of its pieces from the physical dictionary of §1. The entire meta-optimization is a single description-length problem—fit the data with the least total description—whose free variables are, from outermost to innermost, the computational *contract* (which contract the datum lives in), the *width* and *depth* of the network in that contract, and the *weights*. Written as one statement, with

the fixed/variable boundary made explicit:

$$\min_{\substack{\text{activation } \sigma (\Rightarrow \{\Delta_k\}, \alpha) \\ \text{depth } L^*(\mu) \text{ or geodesic } \gamma^* \\ \text{width } \rho^* \text{ (atomic)} \\ \text{weights=saddle of } S}} \left[\underbrace{R[z^{(L)}]}_{\text{fit}} + \underbrace{\lambda \sum_{\ell} \|\rho^{(\ell)}\|_{\text{TV}}}_{\text{width}} + \underbrace{\mu L}_{\text{depth (Route 2)}} \right], \quad (1)$$

subject to the nested stationarity hierarchy (each rung is derived in the section named against it below):

- **Weights** — saddle of the action: forward pass + backprop ($\delta S/\delta z = 0$), the innermost loop, always fixed-graph (§7).
- **Width** — dual certificate $|\eta| \leq 1$ saturating on support: atomic ρ^* , finitely many neurons per layer (§4).
- **Depth** — marginal-value rule $-\partial S^*/\partial L = \mu$ (Route 2), $L^* \sim \mu^{-1/(\alpha+1)}$; or geodesic length minimization (Route 1), depth as gauge (§6).
- **Universality class** — the exponent tower $\{\Delta_k\}$, selected to optimize the depth–complexity trade-off, tuned to criticality $\chi_1 = 1$ where the problem is well posed (§8).

Above all four sits the *kinematic* rung—the choice of contract itself—treated first, in §3, because the notion of a “mode” that width and depth count is undefined until the contract is fixed. The scope of validity of the inner rungs is previewed in Table 2 and established in the sections that follow: convexity, the fixed dual certificate, and exact atomicity are two-layer facts; the saddle-point/RG structure survives to all depths *perturbatively* because it is a local (around-a-point) analysis, whereas the global guarantees required linearity in a single measure, which composition destroys.

Structure	Two layers	Deep ($L \geq 3$)
Variational formulation ($\mathcal{F}$ over measures)	exact	survives
Wasserstein flow, $\mathcal{F}$ as Lyapunov	exact	survives
Gibbs stationarity, path-integral reading	exact	survives (self-consistent)
Joint convexity	holds	<i>lost</i> (multilinear-through- σ)
Fixed-operator dual certificate $\eta = A^*q/\lambda$	holds	<i>lost</i> (A is ρ -dependent)
Exact finite atomicity (cardinality bound)	holds	<i>lost</i>
Saddle = trained net; NTK = propagator; $1/m$ loops	—	holds (perturbative)
Depth-RG criticality, exponent tower	—	holds

Table 2: What transfers from two layers to deep networks. The variational and field-theoretic *structure* survives; the global *guarantees* (convexity, certificate, atomicity) do not.

The rungs are causally ordered. The nested optimizations above are not merely listed from outer to inner; their order is *forced*, and it is the order a physicist imposes on an effective theory. The universality class (which contract, which symmetry group) is the *kinematics*—it fixes the contract and its invariances, and it must be settled first because the very notion of a “mode” is undefined until the contract is chosen. Width and depth are then the *degrees of freedom*: how many modes the fixed contract carries. Training the weights and the primitive parameter α is the *dynamics*: the couplings are solved on that fixed field content. The discretization of the trained mixture into a deployed architecture is the reading-off of *observables*. This is the same sequence as

symmetry-before-dynamics (Noether) and ultraviolet-before-infrared (the renormalization group): kinematics $\rightarrow$ degrees of freedom $\rightarrow$ dynamics $\rightarrow$ observables, with the description-length action $J = R + \mu \Omega$ —in which Ω is the generic complexity penalty, whose concrete instances are the width total-variation norm and the depth count in (1)—minimized *at* each rung rather than constituting a rung of its own. The causal direction is not a stylistic preference but a consistency requirement: a mode count measured before the contract is fixed, or a symmetry quotient applied after the modes are counted, is ill-posed, and (as the implementation companion documents) attempting either produces concrete failures.

2.1 Two corollaries: interpretability by construction, and redundancy reduction as RG-by-measurement

Two capabilities fall out of the same variational structure rather than being added on top of it.

Interpretability is a corollary of MDL selection. When the description-length action selects a sparse, discrete architecture—one primitive per cell, a definite width and depth—the selected object *is* the explanation: there is no separate surrogate model to interrogate, because the quantity that was minimized is exactly the quantity that is read out. This yields three faithful-by-construction tiers. At the level of primitives, the Gibbs weights $\alpha_k \propto e^{-\beta E_k}$ and the clean-solo energies say which primitive each cell chose and how decisively (the participation ratio of α measures peakedness; ties are surfaced rather than broken silently). At the level of input features, an L_0 -gated fit exposes which invariant coordinates carry the signal—but only when the certificate for a *unique* sparse support exists; on a redundant (collinear) basis no unique support exists and the honest output is a reliability flag, not a fabricated ranking. At the level of learned functions, a Kolmogorov–Arnold-network (KAN [4, 1]) radial can be projected onto an analytic dictionary, and here alone a genuine fidelity gap appears: the spline is ground truth and every closed form carries a residual, so the symbolic law is reported as “a fit, not the law,” declining when the residual is too large. The pattern is uniform: the explanation is trustworthy exactly to the extent that the selection problem had a well-posed unique answer, and the framework reports that well-posedness alongside the answer.

Redundancy reduction is the RG “keep the relevant modes” move, by measurement. Symmetry quotients remove redundancy that is known *a priori* from a group; but data carries redundancy no group predicts—correlated, low-variance directions that a coarse-graining should integrate out. The same participation-ratio functional that measures the effective number of primitives, applied one level down to the covariance spectrum of the data, measures the effective number of *modes* d_{eff} ; projecting onto the leading $\lceil d_{\text{eff}} \rceil$ collective coordinates is the renormalization-group step of retaining the relevant operators and discarding the irrelevant ones, now driven by a measured spectral weight rather than a scaling dimension. This is the natural completion of the kinematics rung: the group fixes the *exact* redundancy, and the measured effective dimension fixes the *approximate, data-dependent* redundancy that remains, both before any mode is counted.

From metaphor to a measured flow: the Gaussian information bottleneck [8, 9] as an exact RG. The participation-ratio d_{eff} above is a *static* snapshot—a single number from the covariance of the data alone, with no scale and no notion of relevance. The renormalization group is properly about how such a quantity *flows* as degrees of freedom are integrated out scale by scale, and that flow can be made exact here rather than analogical. For jointly Gaussian (X, Y) the information bottleneck admits a closed-form solution (Chechik–Globerson–Tishby–Weiss): the

optimal compressed representation is a noisy projection onto the eigenvectors of the normalized regression matrix $\Sigma_{X|Y}\Sigma_X^{-1}$ —the canonical-correlation basis—with eigenvalues $\lambda_i \in (0, 1)$, and this coarsening is formally a soft-cutoff non-perturbative RG map with a semigroup structure (Kline–Palmer). The bottleneck tradeoff parameter β is an inverse RG scale: mode i switches on only once β exceeds a critical value $\beta_c(\lambda_i) = 1/(1 - \lambda_i)$, so the effective dimension

$$d_{\text{IB}}(\beta) = \#\{i : \lambda_i < 1 - \frac{1}{\beta}\} \quad (2)$$

is a *staircase* that grows from full compression to full rank through a cascade of structural phase transitions, with each transition located exactly. This is the quantitative content the “keep the relevant modes” metaphor lacked: a flow with pinned transition scales, the most informative modes (smallest λ , largest canonical correlation with Y) surviving to the coarsest scales exactly as relevant operators persist under RG. The same Gaussian-IB spectrum, evaluated on the hidden representation at each layer of a trained network rather than along β , measures the RG reading of depth: on the data examined the top canonical correlation with the target rises monotonically through the layers, so the network is measurably coarse-graining toward the label. The construction is exact only for Gaussian statistics—on non-Gaussian data it is the second-order (canonical-correlation) reading of the bottleneck, not the full nonlinear one—and it is offered as a diagnostic rather than folded into the selection pipeline; but within that scope it replaces an analogy with a measured flow.

One ledger: effective dimension across the coarse-graining axis. The three quantities just named are not three coincidences of a shared word; they are one functional read at successive levels. The participation ratio $\text{PR}(s) = (\sum_i s_i)^2 / \sum_i s_i^2 = 1 / \sum_i p_i^2$ (equal to 1 on a single-mode spectrum and m on a uniform m -mode one) is what counts the effective number of primitives when it eats the mixture weights α , and the effective number of data modes d_{eff} when it eats the covariance eigenvalues—the same map on two different spectra. The Gaussian-IB staircase $d_{\text{IB}}(\beta)$ resolves that same data-mode count against an RG scale, and the real log canonical threshold λ of §9.5 measures the effective dimension of the whole fitted *model*. Stated together these form a single *effective-dimension ledger* along the coarse-graining axis: data modes (unsupervised d_{eff} , or supervised $d_{\text{IB}}(\beta)$), then the primitive mixture ($\text{PR}(\alpha)$), then the model (λ). The unification is exact where it is exact and bounded where it is not: the two kinematic legs are *literally* the same participation-ratio functional, so they share a measurement and not merely a metaphor; λ is a *different* functional (the RLCT, a geometric invariant of the loss-singular set, not a variance ratio) that answers the same question at the model level and caps the ledger at $\lambda \leq k/2$. The one honest caveat, which the ledger states rather than hides, is that the levels are not one commensurable scalar—a count of effective data modes and a count of effective primitives carry different units and are never summed. So the object is one functional read at the kinematic levels plus λ at the model level: the quantitative completion of the “kinematics $\rightarrow$ degrees of freedom” arrow that organizes the whole hierarchy, stated once as a measured quantity rather than as four separate metaphors.

3 Kinematics: the outermost rung, representation dispatch and the discrete–continuous boundary

The consolidated formulation (1) nests four optimizations—weights, width, depth, universality class—all conducted *inside a fixed computational contract*: a definite choice of how the datum is presented to the network (a sequence, a 2D grid, a graph, an unordered set, a set of points with

coordinates). §2 already named this contract as the outermost, *kinematic* rung; here we treat it. It sits *above* all the others, and—as we now show—it is qualitatively different from them.

Why the outer rung is qualitatively different. Inside a contract the primitive is selected by relaxing a discrete choice to a simplex and letting a continuous parameter α flow—the mechanism that the measure-space and Gibbs constructions of the degrees-of-freedom and observable rungs (§4, §9) legitimize, because all primitives in a contract share one interface and their outputs are the *same kind of tensor*, so a convex combination

$$z^{(\ell+1)} = \sum_p \text{softmax}(\alpha)_p f_p(z^{(\ell)}), \quad f_p : \mathcal{T} \rightarrow \mathcal{T}, \quad (3)$$

is a well-defined element of the same tensor space $\mathcal{T}$. Across contracts this fails at the level of types. A sequence core maps $(b, T, c) \rightarrow (b, T, c)$; a volumetric core maps $(b, c, D, H, W) \rightarrow (b, c, D, H, W)$; a graph core maps a variable-node feature matrix with an edge set to itself; a set core maps an unordered collection to itself. There is no common space $\mathcal{T}$ in which their outputs live, hence no convex combination (3) to relax. The outermost selection is therefore *intrinsically discrete*: a dispatch, not a continuous interpolation. The complexity ladder is **continuous within a contract and discrete across contracts**, and the boundary between the two regimes is exactly the boundary of interface compatibility.

Structural dispatch (Level 1). The contract is almost always fixed by the data’s own type, not by a learned signal. Write the datum as a container that either carries an adjacency (edges), a coordinate map (positions), or neither, and is either a fixed-shape array or a variable-size collection. Then the contract is read off structurally:

$$\begin{aligned} \text{edges present} &\Rightarrow \text{graph}, & \text{edges} + \text{positions} &\Rightarrow \text{equivariant}, \\ \text{collection, no edges} &\Rightarrow \text{set}, & \text{fixed array} &\Rightarrow \text{grid family}, \end{aligned} \quad (4)$$

and within the grid family the tensor rank fixes the member (sequence/spatial/volumetric/4D), a flat feature vector being sent to the mutual-information mode-structure test that recovers a latent 1D or 2D lattice. This level is rule-based and certain because the container *is* the answer; it is the coarsest description-length decision, taken before any weights exist.

The one continuous shadow: contract as a hyperprior (Level 2). There is a single genuinely ambiguous case—data carrying coordinates, which may legitimately be modelled as an equivariant graph (geometry used, $\text{SO}(3)$ -equivariant messages), a plain graph (topology only), or a bare point set (no relational structure). Here the “right” contract is not deducible from the container; it depends on whether the target function actually depends on geometry, topology, or neither. Because the contracts cannot be mixed, one cannot relax; the principled substitute is a *bake-off*. Each admissible contract c is fit in isolation on a held-out split and scored by validation risk $\hat{R}_c$, and the winner minimises the same $R + \mu \Omega$ objective as every inner rung,

$$c^* = \arg \min_{c \in \mathcal{C}(\text{data})} \hat{R}_c + \mu_c \Omega_{\text{struct}}(c), \quad \Omega_{\text{struct}}(\text{set}) < \Omega_{\text{struct}}(\text{graph}) < \Omega_{\text{struct}}(\text{equivariant}), \quad (5)$$

where—crucially— $\Omega_{\text{struct}}(c)$ is not an ordinal fiat but a *derived* structural code length: the description length of the scaffolding the contract commits to, per datum. Measuring each contract as

excess over the cheapest (set), for a datum with N units and E edges in d dimensions,

$$\begin{aligned}\Omega_{\text{struct}}(\text{set}) &= 0, \\ \Omega_{\text{struct}}(\text{graph}) &= \log \binom{N(N-1)/2}{E} \approx E \log \frac{N^2}{2E}, \\ \Omega_{\text{struct}}(\text{equiv}) &= \Omega_{\text{struct}}(\text{graph}) + (Nd - \frac{d(d-1)}{2}) \log \frac{1}{\delta},\end{aligned}\tag{6}$$

i.e. the bits to write down the adjacency (graph, the topology) and then the geometry (Nd coordinates at resolution $1/\delta$ minus the $\dim \text{SO}(d)$ gauge the equivariance quotients out). This is strictly monotone up the structure-inclusion lattice and data-dependent, and it is the *structural* richness—the group/interface commitment—not the parametric complexity: the latter would misorder, since an equivariant model can be parameter-cheap yet structurally over-rich for a topological target. The charge is what makes a richer contract earn its keep: among contracts whose risks tie within noise, the cheapest is chosen (a geometry-blind topological target does not pay for $\text{SO}(3)$ -equivariance, because $\Omega_{\text{struct}}(\text{equiv}) \gg \Omega_{\text{struct}}(\text{graph})$ breaks the tie). This is the identical least-description-length logic that governs the inner rungs—the atomic-width certificate, the marginal-value depth rule—now applied one level up, to the choice of interface itself, with a *derived* code length in place of the earlier hand-set ordinal charge. The bake-off is precisely the across-contract analogue of the clean, from-scratch, single-primitive comparison that the within-contract theory uses to defeat co-adaptation: with no shared interface to support a mixed supernet, isolated training is not merely a robustness option but the *only* available comparison.

An amortized surrogate for the argmin. The bake-off (5) evaluates $\hat{R}_c$ by training each contract, which is expensive and, run afresh per dataset, learns nothing across datasets. But c^* is a *function* of the data, and that function is learnable: a cheap descriptor $\phi(\text{data})$ —built from training-free correlations of the target with geometry-, topology-, and set-specific invariants—can predict $\arg \min_c [\hat{R}_c + \mu_c \Omega_{\text{struct}}(c)]$ without computing any $\hat{R}_c$. This is a standard amortization: the optimization (5) is replaced, when a learned predictor is confident, by a forward pass of that predictor, with the bake-off retained as the ground-truth fallback that also supplies fresh training pairs (ϕ, c^*) . The construction changes nothing in the objective— Ω_{struct} is still the structural prior, the label is still the MDL argmin—only *how* the argmin is obtained. Two points are worth recording. The labels must come from an adequately-resolved bake-off: at too small a budget the risk $\hat{R}_{\text{equiv}}$ is not yet converged and the Ω_{struct} charge spuriously prefers the cheaper contract, so a surrogate trained on such labels inherits the error; supplying full-budget labels is what makes the amortization correct. And the descriptor’s transferable coordinates are the *dimensionless* ones (the invariant correlations, each in $[0, 1]$), not the raw extensive counts (nodes, edges), which are not comparable across physical domains—a point that is empirical but has the flavour of a scaling argument. The implementation report documents the realization and its cross-domain validation.

The revised ladder. The consolidated statement (1) thus acquires an outermost, discrete rung:

$$\begin{array}{ccccccc} & & \underbrace{c^*}_{\text{contract (discrete dispatch / MDL bake-off)}} & \succ & \underbrace{\text{universality class}}_{\{\Delta_k\}} & \succ & \underbrace{\text{depth}}_{L^*(\mu)} \\ & & & & & & \\ \succ & \underbrace{\text{width}}_{\rho^* \text{ atomic}} & \succ & \underbrace{\text{primitive}}_{\alpha \text{ simplex}} & \succ & \underbrace{\text{weights}}_{\text{saddle of } S} & \end{array}\tag{7}$$

read from coarse to fine. Everything to the right of the contract is continuous and lives in the variational/field-theoretic account of the preceding sections; the contract itself is the discrete seam

where that account stops, because interface incompatibility forecloses relaxation. Two honest observations follow. First, this outermost rung is *more* principled than the primitive rung immediately below it, not less: the dispatch (5) is a bona fide description-length minimization. (The primitive parameter α one level down, imported from differentiable architecture search, is derived from the same variational principle in §9.) Second, the discreteness is not a defect to be relaxed away but a true feature of the object: the space of computational contracts is a finite set of incompatible categories, and the correct operation on a finite set of incompatible categories is selection, not interpolation.

3.1 Which contract? The group is discovered, not assumed

The dispatch above selects a contract by description length, but it treats the candidate contracts as given. A contract is, at bottom, the natural equivariant architecture for a symmetry group G ; so the sharper question is which G the data respects. This is answerable by the same variational hygiene used throughout: a group is admissible exactly when its invariants explain the target, and the correct G is the one whose invariant structure the data actually carries.

The invariance test. For a candidate group with metric g , the quantities $\langle p_i, p_j \rangle_g = p_i^\top g p_j$ are exactly invariant under every transformation L with $L^\top g L = g$. One fits the target from these invariants and compares against the frame-sensitive raw coordinates; the metric that fits is the group. Crucially, model selection here must be controlled not by in-sample fit—a sufficiently expressive invariant descriptor can overfit frame-dependent signal—but by *stability*: the fit must survive the group’s own transformations. Applying a random $L = \exp(\sum_k t_k A_k)$ to a held-out split leaves a genuinely G -invariant target’s fit unchanged and collapses a spurious one. This is the discrete analogue of a Ward identity: the admissible symmetry is the one under whose action the fitted response is stationary.

Discovering the metric. The metric itself need not be assumed. A generator A preserves g iff $A^\top g + g A = 0$; given generators, the invariant metric is the symmetric g in the null space of this linear constraint, and its eigenvalue signature (p, q) —taken up to an overall sign gauge—names the pseudo-orthogonal group $O(p, q)$. This is the identical null-space computation that produces an equivariant layer, with the metric rather than the layer as the unknown, and it recovers arbitrary signatures (Euclidean, Lorentz, and split forms such as $O(2, 2)$) exactly. When the target is a metric norm $y = \langle g, ss^\top \rangle$, the recovery is even simpler—linear in g , hence an exact least squares—which on real four-vector data returns the Minkowski metric with unit R^2 .

A lattice of invariant structures. Not every symmetry preserves a symmetric bilinear form. The metric family $O(p, q)$ preserves $g = g^\top$; the symplectic family $Sp(2n)$ preserves a skew form $\omega = -\omega^\top$; the special-linear family $SL(n)$ preserves the alternating volume form; the unitary family $U(n)$ preserves a Hermitian form, equivalently a Euclidean metric together with a compatible complex structure J ; and the conformal family $\text{Conf}(d) = O(d + 1, 1)$ preserves the light-cone metric on the null-cone lift. Each is detected by regressing the target against the corresponding algebraic invariant—symmetric, skew, alternating, phase-invariant, or lifted—and each regression carries its own fit gate, so a target that carries no such invariant yields no group. Selecting among them is again description-length minimization in disguise: prefer the richest structure the data admits, and decline when it admits none. The outermost rung of the dispatch is thus not merely “which container” but “which group,” and the answer is read off from the invariant the target is willing to be written in.

4 Degrees of freedom I: width at fixed architecture (Scenario 0)

4.1 Convexification by lifting to a measure

The starting difficulty is that the empirical risk of a finite-width network is a non-convex function of the parameters $\{\Theta_j\}$. The lift that removes this is to replace the empirical measure $\frac{1}{m} \sum_j \delta_{\Theta_j}$ by an arbitrary measure $\rho \in \mathcal{M}(\Theta)$ and write the network as a *linear* functional of ρ :

$$f_\rho(x) = (A\rho)(x) = \int_{\Theta} \phi(x; \Theta) d\rho(\Theta), \quad \phi(x; \Theta) = a \sigma(u^\top x), \quad (8)$$

where $A : \mathcal{M}(\Theta) \rightarrow L^2$ is a bounded linear operator. Because f_ρ is linear in ρ , the squared-error risk $\rho \mapsto \frac{1}{2} \|A\rho - y\|_{L^2}^2$ is *convex*. This is the measure-space (mean-field) convexification: non-convexity in $\mathbb{R}^d$ becomes convexity in $\mathcal{P}(\Theta)$.

To promote *sparsity* of the representation—i.e. to prefer few effective neurons—we regularize by the total-variation (atomic) norm $\|\rho\|_{\text{TV}}$, giving the total-variation-regularized free energy $\mathcal{F}_{\text{TV}}$:

$$\boxed{\mathcal{F}_{\text{TV}}[\rho] = \frac{1}{2} \|A\rho - y\|_{L^2}^2 + \lambda \|\rho\|_{\text{TV}}.} \quad (9)$$

This is the Beurling–LASSO / atomic-norm problem. Its role in the MDL reading is that λ is the description-length price of one atom (neuron); larger λ buys a coarser, cheaper model. (In §4.3 the same free energy $\mathcal{F}$ appears with an entropic penalty in place of the TV one; we write $\mathcal{F}_{\text{TV}}$ and $\mathcal{F}$ for the two regularizations of the one objective.)

4.2 The dual certificate and finite atomicity

The optimality condition for (9) is best stated through the *dual certificate*. Let $q^* = y - A\rho^*$ be the optimal residual and define

$$\eta = \frac{1}{\lambda} A^* q^* \in C(\Theta), \quad (A^* q)(\Theta) = \langle q, \phi(\cdot; \Theta) \rangle_{L^2}. \quad (10)$$

The Karush–Kuhn–Tucker conditions for the convex program (9) are

$$\boxed{|\eta(\Theta)| \leq 1 \quad \forall \Theta, \quad \eta(\Theta) = \text{sign}(\rho^*(\Theta)) \text{ on } \text{supp}(\rho^*), \quad |\eta(\Theta)| < 1 \text{ off support.}} \quad (11)$$

Geometrically, the support of the optimal measure sits exactly at the *contact points* where the certificate saturates, $\{|\eta| = 1, \nabla_{\Theta} \eta = 0\}$.

Proposition 1 (Finite atomicity). *Under (11), if the contact set $\{|\eta| = 1\}$ is finite, the minimizer ρ^* of (9) is atomic with finitely many atoms; the minimal network is provably finite-width.*

This is the representer theorem for TV-regularized measure recovery (extreme-point / Krein–Milman characterization of the TV ball). It is the crown jewel of Scenario 0: the *optimal number of neurons emerges from convex geometry* rather than being fixed by hand.

Remark 1 (A subtlety exposed in practice). *Greedy neuron insertion with ordinary least-squares refitting is not the TV optimum: OLS forces the residual orthogonal to the active atoms, which drives $\eta \rightarrow 0$ on support and violates (11). The certificate is what exposes this; the correct frontier is obtained by conditional-gradient / column-generation against the TV penalty. The true support is typically large, not artificially sparse.*

Remark 2 (The certificate fixes the price scale: λ is calibrated, not free). *The certificate also settles what value the width price λ should take. Since $\eta = A^*q^*/\lambda$ must satisfy $|\eta| \leq 1$ with equality on the support, the largest price at which any atom still enters is $\lambda^* = \|A^*q\|_\infty$, the peak correlation of the residual with the candidate neurons; above it the certificate is slack and every atom is pruned, below it new atoms enter. Evaluated at the null model $q = y$ this gives a data-driven scale $\lambda_0^* = \|A^*y\|_\infty$ (over a dictionary of candidate neurons), so the meaningful sweep of λ is a fraction of λ_0^* rather than an arbitrary grid. This does not hand a single universal scalar— λ^* scales with the residual and, for a deep gated network, the effective per-neuron price is a depth- and conditioning-dependent fraction of the single-atom value—but it fixes the scale per dataset from a measured quantity, turning λ from a free knob into a calibrated one. In the implementation this is the certificate- λ used to set the operating range of the generalized-area gate.*

4.3 Dynamics: Wasserstein gradient flow of a free energy

Replacing the TV penalty by an entropic one gives the free-energy form, with KL the Kullback–Leibler divergence (relative entropy),

$$\mathcal{F}[\rho] = R[\rho] + \frac{1}{\beta} \text{KL}(\rho \| \pi), \quad (12)$$

with prior π and inverse temperature β . The stationarity condition $\delta\mathcal{F}/\delta\rho = \text{const}$ yields the self-consistent *Gibbs law*

$$\frac{d\rho^*}{d\pi}(\Theta) \propto \exp(-\beta \Psi_{\rho^*}(\Theta)), \quad \Psi_\rho = \frac{\delta R}{\delta \rho}, \quad (13)$$

and the training dynamics are the Wasserstein gradient flow (McKean–Vlasov / mean-field Langevin equation)

$$\partial_t \rho = \nabla \cdot \left(\rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right) + \frac{1}{\beta} \Delta \rho, \quad (14)$$

for which $\mathcal{F}$ is a Lyapunov function. Thus “training as descent toward least complexity” is literal: the flow monotonically decreases the free energy toward the Gibbs stationary measure. Equation (13) is the seed from which the *architecture* Gibbs measure of the outer loop later grows.

5 The obstruction at depth: where convexity dies

The convexity and the fixed-operator certificate are properties of the *two-layer* lift. It is essential to see precisely how they fail at three layers, because the failure dictates the physics of the deep formulation.

Write a three-layer network in mean-field scaling,

$$f(x) = \frac{1}{m_2} \sum_{k=1}^{m_2} a_k \sigma \left(\underbrace{\frac{1}{m_1} \sum_{j=1}^{m_1} w_{kj} \sigma(u_j^\top x)}_{\text{second-layer preactivation}} \right). \quad (15)$$

Lifting each layer to a measure ρ_1, ρ_2 produces a *nested* integral,

$$f(x) = \int a \sigma \left(\int w(\Theta_2, \Theta_1) \sigma(u(\Theta_1)^\top x) d\rho_1(\Theta_1) \right) d\rho_2(\Theta_2). \quad (16)$$

The inner integral sits *inside* the nonlinearity σ , so f is linear in the outer measure ρ_2 but *nonlinear* in the inner ρ_1 . The map $(\rho_1, \rho_2) \mapsto f$ is affine in each argument separately but not jointly, and the

functional $\mathcal{F}[\rho_1, \rho_2] = R + \lambda_1 \|\rho_1\|_{\text{TV}} + \lambda_2 \|\rho_2\|_{\text{TV}}$ is convex in each argument with the other fixed but *not jointly convex*. Global optimality collapses to local/critical-point statements, and because $A = A[\rho_1]$ now depends on the measure, there is no fixed adjoint A^* and hence no fixed certificate $\eta = A^*q/\lambda$; (11) weakens to a self-consistent fixed-point statement. Exact atomicity (Prop. 1) is likewise lost, as the extreme-point argument required linearity in a single measure. What *survives* is the variational/free-energy *formulation* itself: $\mathcal{F}$ over path measures is well defined, the Wasserstein flow still has $\mathcal{F}$ as a Lyapunov function, and stationary measures still exist.

6 Degrees of freedom II: depth as an optimized variable (Scenario 1)

We now promote depth from a fixed parameter to an optimized variable. Classical mechanics offers exactly two ways to make “elapsed time” a variable—fixed-endpoint variation with a time cost, and free-endpoint (Maupertuis) variation—and depth-as-complexity inherits the fork. They are Legendre-dual.

6.1 Route 2: priced depth (canonical ensemble)

Keep depth a discrete, countable resource and charge it with a chemical potential μ . The Route-2 objective adds a linear depth price to the action, $\mathcal{J}_2[z, L] = S_L[z] + \mu L$, and the corresponding partition function sums over depth:

$$\mathcal{Z} = \sum_{L=1}^{\infty} e^{-\beta \mu L} \int \mathcal{D}z e^{-\beta S_L[z]} = \sum_L e^{-\beta \mathcal{J}_2[z^*, L]} (1 + \text{fluctuations}), \quad \mathcal{J}_2[z^*, L] = S_L^* + \mu L. \quad (17)$$

This is a *grand-canonical ensemble in depth*: μ is conjugate to layer number exactly as a chemical potential is conjugate to particle number. There are now two stationarity conditions: the usual field equation (27) at fixed L , and a new *depth condition*

$$\boxed{-\left. \frac{\partial S_L^*}{\partial L} \right|_{L^*} = \mu.} \quad (18)$$

In words: *add layers until the marginal reduction in action per layer falls to the price μ* . This is the depth analogue of the width certificate: neurons were added until the correlation potential fell below λ ; layers are added until the marginal action-gradient falls below μ .

Remark 3 (Discreteness). *Because $L \in \mathbb{Z}$, (18) is a discrete stopping rule $S_L^* - S_{L+1}^* \geq \mu$ —a free-boundary / variational-inequality statement, not a smooth Euler–Lagrange equation. This is the same discreteness obstruction as the $\|\theta\|_0$ penalty, now in the depth direction; the continuum limit (Route 1) is what smooths it.*

Optimal depth from the exponent. The RG analysis computes the left-hand side of (18). At criticality the marginal reduction from one more layer is dominated by the marginal mode (33): the irrelevant modes $k \geq 2$ have finite mode-specific correlation lengths $\xi_k = -1/\log |\Delta_k|$ (the analogue of (32) for the k -th eigenvalue Δ_k) and contribute exponentially decaying terms $e^{-\ell/\xi_k}$ (they stop paying after $O(\xi_k)$ layers), while the marginal mode contributes a power-law tail,

$$-\frac{\partial S_\ell^*}{\partial \ell} \sim c_1 \ell^{-(\alpha+1)} + \sum_{k \geq 2} c_k e^{-\ell/\xi_k}, \quad (19)$$

with mode amplitudes $c_1, c_k > 0$ set by the projection of the loss onto each RG eigenmode. Equating the surviving power law to μ gives the optimal depth

$$\boxed{L^* \sim \mu^{-1/(\alpha+1)}} \quad (\text{smooth } \alpha = 1 : \mu^{-1/2}; \quad \text{ReLU } \alpha = 2 : \mu^{-1/3}). \quad (20)$$

Off criticality the marginal value decays exponentially, so L^* saturates at the finite correlation length ξ regardless of μ : *optimal depth becomes a tunable power law of the price only on the critical line.* This is the precise sense in which the depth chemical potential “vanishes at criticality.”

Remark 4 (Generalized area: variable width-per-layer with emergent depth). *The width penalty $\lambda \sum_\ell \|\rho^{(\ell)}\|_{\text{TV}}$ is already a per-layer sum of atomic norms, and $\|\rho^{(\ell)}\|_{\text{TV}} \sim m_\ell$ counts the neurons kept in layer ℓ . Collapsing to a common width $m_\ell \equiv m$ gives $\lambda m L = \lambda \cdot (\text{area})$, the uniform width-times-depth cost; the uncollapsed penalty is the **generalized area** $\sum_\ell m_\ell$, the total neuron count. Pricing this single quantity, $R + \lambda \sum_\ell m_\ell$, makes both the per-layer width and the depth outputs of one functional: depth is emergent, $L = \#\{\ell : m_\ell > 0\}$, since a closed layer costs zero neurons and zero depth, absorbing the μL term into the same neuron count (the $\mu \rightarrow 0$ /critical face, up to an optional separate per-active-layer charge). This is the most general form of the size problem, and it is realized by per-neuron L_0 gates whose expected open-count is a differentiable m_ℓ ; empirically it recovers variable per-layer widths and closes redundant layers (emergent depth) on real data.*

6.2 Route 1: geometric depth (microcanonical ensemble)

In the continuum limit, $\sum_\ell \rightarrow \int_0^T dt$ and the network is a flow $\dot{z} = g(z, \theta(t))$. Now the number of layers has no invariant meaning—the Route-1 objective $\mathcal{J}_1$ (the continuum counterpart of $\mathcal{J}_2$) should be invariant under monotone reparameterization $t \mapsto \tau(t)$. Demanding reparameterization invariance of the complexity term forces its integrand to be homogeneous of degree one in $\dot{z}$, hence a Finsler/Riemannian *length*:

$$\boxed{\mathcal{J}_1[\gamma] = R(z(T)) + \int_0^T \|\dot{z}(t)\|_{M(z(t))} dt, \quad \|\dot{z}\|_M = \sqrt{\dot{z}^\top M(z) \dot{z}}.} \quad (21)$$

The Euler–Lagrange equation is the *geodesic equation* in the learned metric $M(z)$,

$$\ddot{z}^i + \Gamma_{jk}^i(z) \dot{z}^j \dot{z}^k = 0, \quad \Gamma_{jk}^i = \frac{1}{2} M^{il} (\partial_j M_{lk} + \partial_k M_{lj} - \partial_l M_{jk}), \quad (22)$$

with transversality $M\dot{z} \parallel \nabla R$ at the endpoint. The least-complexity network is the *shortest curve* from the input manifold to the loss-minimizing terminal set; complexity is arc length; depth is the affine parameter (gauge).

Depth-freedom from the exponent. The metric component along the marginal direction scales, in the continuum limit of (33), as $M_{\text{marg}}(t) \sim t^{-2\alpha}$, so the accumulated arc length in that direction is

$$\text{Length}_{\text{marg}} = \int_0^T \sqrt{M_{\text{marg}}} dt \sim \int_0^T t^{-\alpha} dt, \quad (23)$$

whose convergence is decided by α :

$$\int_0^T t^{-\alpha} dt \longrightarrow \begin{cases} \text{converges (depth asymptotically free),} & \alpha > 1 \text{ (ReLU),} \\ \log T \text{ (logarithmic cost per decade),} & \alpha = 1 \text{ (smooth),} \\ \text{diverges as a power,} & \alpha < 1. \end{cases} \quad (24)$$

This predicts that a ReLU continuum-depth model can be arbitrarily deep at finite complexity cost, whereas a smooth-activation model pays a logarithm per decade of depth.

6.3 Legendre duality of the two routes

Routes 1 and 2 are related exactly as the free-time and fixed-time formulations of mechanics: the chemical potential μ (price of a layer) is conjugate to the depth/time T , as energy is conjugate to time. Concretely $\mu = -\partial S^*/\partial L$ is the slope of the depth–loss curve and the geodesic length is its Legendre transform. Sweeping μ in Route 2 traces the same fit–depth Pareto frontier as choosing where to stop along the Route 1 geodesic.

7 Dynamics: the path-integral formulation of the trained network

The clean way to carry the variational structure to arbitrary depth is to abandon per-layer measures in favor of a single measure over *configurations of the whole network*, and to organize the calculation as a path integral.

7.1 Partition function over preactivation fields

Let $z_i^{(\ell)}(x)$ be the preactivation of unit i in layer ℓ . The forward map is the constraint

$$z_i^{(\ell+1)}(x) = \frac{1}{\sqrt{m}} \sum_j W_{ij}^{(\ell)} \sigma(z_j^{(\ell)}(x)), \quad W_{ij}^{(\ell)} \sim \mathcal{N}(0, \sigma_w^2). \quad (25)$$

Integrating out the raw Gaussian weights $W_{ij}^{(\ell)}$ and appending the data/complexity term R yields a path integral over the preactivation fields,

$$\mathcal{Z} = \int \mathcal{D}z \, e^{-\beta S[z]}, \quad S[z] = R[z^{(L)}] + \frac{1}{2} \sum_{\ell} \|z^{(\ell+1)} - \mathcal{W}^{(\ell)} \sigma(z^{(\ell)})\|_2^2, \quad (26)$$

where the script symbol $\mathcal{W}^{(\ell)}$ denotes the *effective* (integrated-out) coupling of layer ℓ —distinct from the raw random weight $W_{ij}^{(\ell)}$ of (25), which no longer appears once integrated out. The norm $\|\cdot\|_2$ is the Euclidean norm on the layer’s unit index. The crucial structural fact is that the action is *local in depth*: $S = \sum_{\ell} \mathcal{L}^{(\ell)}(z^{(\ell)}, z^{(\ell+1)})$, nearest-neighbor in the “time” index ℓ , with σ the interaction vertex. This makes the network a one-dimensional field theory in the depth direction, with a transfer-operator (time-ordered) structure.

7.2 Saddle point: the trained network and the neural ODE

The stationary configuration $\delta S/\delta z = 0$ is, by depth-locality, a two-point boundary value problem:

$$\begin{cases} \text{forward: } z^{(\ell+1)} = \mathcal{W}^{(\ell)} \sigma(z^{(\ell)}), & (\text{input BC at } \ell = 0) \\ \text{backward: } p^{(\ell)} = (\mathcal{W}^{(\ell)})^\top [\sigma'(z^{(\ell)}) \odot p^{(\ell+1)}], & (\text{loss BC at } \ell = L) \end{cases} \quad (27)$$

with costate $p^{(\ell)} = \delta S/\delta z^{(\ell)}$. These are exactly *forward propagation and backpropagation*, read as the forward/adjoint legs of a discrete least-action problem with the loss as terminal condition. In the continuum-depth limit $\ell/L \rightarrow t \in [0, 1]$,

$$\boxed{\dot{z}(t) = g(z(t), \theta(t)), \quad -\dot{p}(t) = (\partial_z g)^\top p(t), \quad p(1) = \nabla_z R.} \quad (28)$$

Here g is the continuum-limit vector field (the limit of the discrete update $z^{(\ell+1)} - z^{(\ell)} = \mathcal{W}^{(\ell)} \sigma(z^{(\ell)}) - z^{(\ell)}$), while $\theta(t)$ are the continuum-depth parameters and $p(t)$ is the continuum

costate (the limit of $p^{(\ell)}$). The first equation is the *neural ODE*; the pair is Pontryagin’s maximum principle, with the adjoint ODE being continuous-depth backpropagation. The classical trajectory of the complexity path integral is the neural-ODE flow.

7.3 Gaussian fluctuations: the NTK as propagator; loops as $1/m$

Expanding $z = z^* + \delta z$ to second order gives $S = S[z^*] + \frac{1}{2} \delta z^\top \mathbb{H} \delta z + O(\delta z^3)$ with Hessian $\mathbb{H} = \delta^2 S / \delta z^2|_{z^*}$. The Gaussian integral yields the propagator

$$\langle \delta z \delta z \rangle = \frac{1}{\beta} \mathbb{H}^{-1} =: \frac{1}{\beta} G, \quad (29)$$

whose block-tridiagonal (nearest-neighbor) structure inverts into exactly the recursive kernel of infinite-width theory: the *neural tangent kernel*. Concretely the layerwise recursion reproduces the standard NTK / neural-network-Gaussian-process (NNGP) relations,

$$\Theta^{(\ell+1)} = \Theta^{(\ell)} \odot \dot{\Sigma}^{(\ell+1)} + \Sigma^{(\ell+1)}, \quad (30)$$

with preactivation covariance $\Sigma^{(\ell)}$ and derivative kernel $\dot{\Sigma}^{(\ell)}$. Thus *saddle* = *trained network*, *propagator* = *NTK*; the lazy/kernel regime is the free-field limit. Cubic and higher terms are interaction vertices, and power counting in width shows the connected n -point functions scale as $\langle \delta z^n \rangle_c = O(m^{-(n/2-1)})$: the tree level ($m = \infty$) is the free NTK theory and each loop costs $1/m$. Feature learning and NTK drift are the leading $1/m$ (four-point vertex) corrections, controlled by the depth-to-width ratio L/m ; the loop expansion is well behaved precisely when $L/m \ll 1$.

Remark 5 (Two scalings, two saddles). *Under NTK scaling ($1/\sqrt{m}$) the saddle is the network at initialization and the kernel is frozen at leading order. Under mean-field scaling ($1/m$) the saddle itself moves and the leading-order dynamics are the nonlinear Wasserstein flow of §4.3; the “free theory” is already interacting. The MDL/TV functional, with its per-layer penalties, lives naturally in the mean-field scaling, which is why its leading order is the interacting flow and the certificate/atomicity structure appears already at tree level.*

8 Criticality: the renormalization group of the depth map

The layer-to-layer covariance recursion (30) is a renormalization-group map in the depth direction; its fixed-point structure controls whether signal propagates. For a tanh-type network the single relevant quantity is the slope of the correlation map,

$$\chi_1 = \sigma_w^2 \mathbb{E}[\sigma'(z)^2], \quad (31)$$

which classifies the phase: $\chi_1 < 1$ *ordered* (correlations contract, gradients vanish), $\chi_1 > 1$ *chaotic* (correlations expand, gradients explode), and $\chi_1 = 1$ *critical* (marginal propagation). The correlation length is

$$\xi = -\frac{1}{\log \chi_1}, \quad (32)$$

diverging at criticality, where ξ is the correlation length of the marginal mode and $\epsilon^{(\ell)}$ below denotes a perturbation to the layer- ℓ preactivation correlation. Away from criticality perturbations decay exponentially, $\epsilon^{(\ell)} \sim e^{-\ell/\xi}$; at criticality the marginal mode decays only algebraically,

$$1 - c^{(\ell)} \sim \ell^{-\alpha}, \quad (33)$$

where $c^{(\ell)} \in [0, 1]$ is the depth- ℓ correlation (normalized overlap) between two nearby inputs propagated to layer ℓ , so $c^{(\ell)} \rightarrow 1$ is the fixed point and $1 - c^{(\ell)}$ its approach; this defines the critical exponent α . Using the Mehler/Hermite spectrum of the activation, the dual activation is $\hat{\sigma}(\rho) = \sum_n a_n^2 \rho^n$ with a_n the Hermite coefficients of σ , and this fixes both the tower of scaling dimensions $\{\Delta_k\}$ (the eigenvalues of the linearized RG map) and the marginal exponent α : $\alpha = 1$ for smooth activations and $\alpha = 2$ for ReLU. The tower $\{\Delta_k\}$ is the universal data that will make the depth objectives computable in §6; it is the same tower referenced in the consolidated objective (1).

Remark 6 (Status of criticality as a lever). *As an initialization prescription, criticality gives the textbook clean gradient-propagation behavior (ordered $\rightarrow$ vanishing, chaotic $\rightarrow$ exploding, critical $\rightarrow O(1)$). As a trainability lever, however, it is largely subsumed by normalization in practice. Its indispensable role in these notes is theoretical: it is the locus where the depth objectives below become nontrivial.*

9 Observables: deriving the architecture parameter

The one component the preceding account imported rather than derived is the primitive-selection parameter α : within a contract, the mixture $z^{(\ell+1)} = \sum_p \text{softmax}(\alpha)_p f_p(z^{(\ell)})$ is trained by gradient descent on α and discretized by arg max, a recipe taken wholesale from differentiable architecture search. We now show this is not a foreign object but the outer Gibbs law that Eq. (13) already anticipated, and that DARTS (differentiable architecture search [3]) is its gradient flow.

9.1 The free energy on the primitive simplex

Place on the primitive simplex $\Delta = \{\alpha \in \mathbb{R}_{\geq 0}^P : \sum_p \alpha_p = 1\}$ the same free energy that governs the neuron measure, now with a discrete index:

$$\mathcal{F}_\alpha(\alpha) = \langle \alpha, \Psi \rangle - \frac{1}{\beta} H(\alpha), \quad H(\alpha) = - \sum_p \alpha_p \log \alpha_p, \quad (34)$$

where Ψ_p is the marginal risk (description-length cost) of primitive p and β is the same inverse temperature as $\mathcal{F} = R + \frac{1}{\beta} \text{KL}$. This is $R + \frac{1}{\beta}(\text{rate})$ read on the lattice of primitives: $\langle \alpha, \Psi \rangle$ is the expected fit cost of the mixture and $-H$ is its description length (how concentrated, hence how committed, the choice is).

9.2 Gibbs form, replicator dynamics, and the DARTS correspondence

The stationarity of (34) on the interior of Δ (Lagrange condition $\nabla \mathcal{F}_\alpha \parallel \mathbf{1}$) gives at once the *Gibbs law*

$$\boxed{\alpha_p \propto \exp(-\beta \Psi_p)} \quad (35)$$

which is exactly Eq. (13), $d\rho^*/d\pi \propto e^{-\beta\Psi}$, transported from the neuron measure to the primitive simplex—the “architecture Gibbs measure of the outer loop” promised there. The natural gradient flow of (34) in the Fisher–Rao (Shahshahani) metric of the simplex is the *replicator equation*

$$\dot{\alpha}_p = -[J \nabla \mathcal{F}_\alpha]_p = \alpha_p (-\Psi_p + \langle \alpha, \Psi \rangle) + \frac{1}{\beta} \alpha_p (-\log \alpha_p + H), \quad J = \text{diag}(\alpha) - \alpha \alpha^\top, \quad (36)$$

i.e. selection dynamics with fitness $f_p = -\Psi_p$, whose fixed point is (35). The DARTS update—gradient descent on the softmax logits—is, in continuous time, $\dot{\alpha} = -J^2 \nabla \mathcal{F}_\alpha$: the replicator field

(36) pushed once more through the positive-semidefinite map J . Since $\ker J = \text{span}(\mathbf{1})$ is exactly the simplex normal, J changes only the metric, not the critical set: DARTS shares the replicator flow’s entire stationary set and its interior Gibbs equilibrium (for $P = 2$ the extra factor is the positive scalar $2\alpha_1\alpha_2$, so the two flows coincide up to a time reparameterization). *The bolt-on is thus the gradient flow of the outer Gibbs measure*, in a mildly preconditioned metric.

9.3 One temperature, three regimes; and which energy is forced

The single knob β organizes the selection exactly as μ organizes depth and λ organizes width. Rewriting (34) as $\langle \alpha, \Psi \rangle + \frac{1}{\beta}(-H)$ exhibits $1/\beta$ as the *price of committing*: at small β the measure is broad (a hedged mixture, “buy fit with capacity”), at large β it sharpens, and as $\beta \rightarrow \infty$ it concentrates on $\arg \min_p \Psi_p$ —the hard, single-primitive selection, the primitive-lattice analogue of the atomic-width dual certificate. Sweeping β traces a fit-versus-commitment Pareto frontier just as sweeping μ traces the fit-versus-depth frontier.

Finally, (34) *forces* the identity of the energy. The variational principle does not merely permit Ψ_p ; it requires the *marginal* (solo) risk of primitive p —its description length as a standalone hypothesis—not the mixture gradient $\partial R/\partial \alpha_p$ that plain DARTS differentiates. The two differ precisely by co-adaptation: the mixture gradient credits a primitive for fitting the residual its neighbours leave, so a parameter-free branch can win a flat mixture. Because the derived energy is intrinsically a solo quantity, the derived selection is co-adaptation-robust *by construction*, not by an added ablation heuristic. Thus the same step that derives α also resolves the selection pathology the within-architecture theory left open: the α -derivation and the co-adaptation fix are one result.

9.4 One free-energy form, three temperatures

The inverse temperature β of (35) is not the only one in the framework, and it is worth being precise about how the several β ’s relate, because a careless reader will suspect a single knob wearing three hats. They do share a single *form*. Every priced read-out in the package is a Gibbs free energy of one shape,

$$F(\rho) = \langle \rho, E \rangle - \frac{1}{\beta} H(\rho), \quad (37)$$

a mean energy against an entropy at an inverse temperature; the architecture free energy (34) is the instance on the primitive simplex, and the same shape recurs at two other levels. But the three instances carry three *different* temperatures, and—this is the substantive point—they are deliberately decoupled, not one temperature.

At the level of the **weights**, the singular free energy of §9.5 is $F_n = nL(w) + \lambda \log n$, whose Gibbs form runs the posterior at $\beta_W = 1/\log n$. This is not a free knob: only that value makes $n\beta_W(\mathbb{E}_{\text{post}}[L] - L^*)$ estimate the real log canonical threshold [10, 2], so β_W is pinned by singular-learning theory. At the level of the **primitives**, (37) with $E = \Psi_p$ on the simplex runs at $\beta_A = \beta$, the genuine free sharpness knob of the previous subsection (small β_A hedges, large β_A commits). At the level of the **contracts**, the evidence expansion of §9.5 combines a per-datum negative log-likelihood with a structural code length, both already in nats, at $\beta_C = 1$: an NLL against a description length admits no free exchange rate (nats against nats), so β_C is unit-fixed. The derivative read-outs of this one form are then the three objects the framework already computes: $\lambda = dF_n/d \log n$ at the weights, the derived α at the primitives, and the contract posterior (whose MAP is the priced- J winner) at the contracts.

The temptation is to “unify” by equating them—setting the readout temperature to the weight temperature, $\beta_A := \beta_W = 1/\log n$. That is a mistake, and seeing why sharpens the picture. At a representative $n \sim 10^3$, $1/\log n \approx 0.14$ whereas a useful β_A is order 10; equating them drops

the readout temperature by nearly two orders of magnitude and flattens α to nearly uniform, destroying the selection the observable exists to make. Nor is there a single partition function whose marginals are all three: the energies live on different spaces and in different units— nL is $O(n)$ nats on weight space, Ψ_p is an $O(1)$ dimensionless fit metric on the simplex, and the contract energy is $O(n)$ nats on the contract set—so no one $Z_n(\beta)$ produces all three. The honest statement is therefore one free-energy *form* instantiated at three coarse-grainings, each with a temperature pinned by its own principle (theory-fixed β_W , free β_A , unit-fixed β_C)—three deliberately decoupled temperatures, not one. This closes a latent notational double-use rather than papering over it, and it is the thermodynamic backbone from which the singular complexity and its developmental and response read-outs are all read as derivatives of the *same* object.

9.5 Mixtures: the dense blend collapses to DARTS, but can be priced

The derivation selects, at $\beta \rightarrow \infty$, a single primitive. Does it extend to a genuine *mixture*—a deployed map $\sum_p \alpha_p f_p$ that can strictly beat every vertex? Here the variational structure draws a sharp and honest line. Over a *frozen* basis $\{f_p\}$ the blended risk $R(\sum_p \alpha_p f_p)$ is convex on the simplex, with a dual certificate identical in form to the width certificate (11)—support primitives share a common marginal correlation with the residual, off-support primitives correlate no better—so the optimal mixture, and its finite support, are as well-characterised as the atomic width. But a frozen basis is exactly post-hoc *ensembling* (a bag of separately trained models), which sits outside the single-network discipline. For a *live* in-layer mixture the basis co-adapts: α and the weights θ minimise the *same* non-convex loss, $\partial R / \partial \alpha_p$ is the gradient of that very loss, and the entropic-mirror-descent stationarity $\alpha_p \propto e^{-\beta \partial R / \partial \alpha_p}$ is therefore a *reparametrisation of DARTS*, not an independent principle. The clean certificate had teeth in Scenario 0 precisely because the two-layer lift made the problem convex in a *single* measure; composition destroys that here just as it does at depth (§5).

What remains, and is fully in keeping with the priced rungs of (1), is to *price* the mixture instead of deriving it. The description length of a mixture over P primitives is, by the two-part code, $\sim k \log P + k \cdot (\text{weight bits})$ with $k = \|\alpha\|_0$ the active count—linear in the support size, exactly as the width penalty is linear in the neuron count. A smooth surrogate for $\|\alpha\|_0$ is the inverse participation ratio $\text{IPR}(\alpha) = 1 / \sum_p \alpha_p^2 \in [1, P]$, so the priced objective

$$\mathcal{J}_{\text{mix}} = R\left(\sum_p \alpha_p f_p\right) + \mu \Omega(\alpha), \quad \Omega(\alpha) = -\sum_p \alpha_p^2 \quad (\text{concentration reward, } \downarrow \text{IPR}), \quad (38)$$

is the primitive-dictionary analogue of the width atomic penalty: a primitive stays in the mixture only while its marginal risk reduction exceeds the price μ , the same conditional-gradient stopping rule as the width certificate (neurons) and the depth marginal-value rule (layers), now on the primitive lattice. Sweeping μ traces an accuracy-versus-effective-primitive-count frontier; a single primitive is the μ -large corner and a genuine mixture the interior, the data deciding which by where its frontier knees. This is the mixture extension the framework actually supports: not a richer derivation of the dense blend—which reduces to DARTS—but a description-length price with a certificate-flavoured reading, placed on the same ladder as width and depth.

For future reference it is important to record what in this construction is novel and what is a synthesis of established results, and—separately—which predictions have been tested.

Provenance. Every load-bearing component is established, frequently foundational, literature: the measure-space convexification and Wasserstein/mean-field dynamics (Mei–Montanari–Nguyen; Chizat–Bach; Rotskoff–Vanden-Eijnden; Sirignano–Spiliopoulos); TV/atomic-norm sparsity with

finite-atom certificates (Bach; Chizat; Parhi–Nowak; Pilanci–Ergen); the $1/m$ effective field theory with NTK-as-propagator (Roberts–Yaida–Hanin; Halverson et al.; the auxiliary-field/loop treatments); depth-RG criticality and correlation-length divergence (Poole et al.; Schoenholz et al.); critical exponents and activation universality classes; the Hermite/Mehler dual-activation spectrum and its fixed-point dynamics (Daniely et al.; Joudaki et al.); and depth-as-optimal-control with regularization-induced depth bounds (E; Benning et al.; Faulwasser et al. turnpike theory). The value of the construction is as a *single internally consistent arc* connecting these programs, not as new results at the component level.

Tested predictions. The one component with a claim to modest novelty is the closed-form coupling $L^* \sim \mu^{-1/(\alpha+1)}$ (Route 2) and its Route 1 counterpart (24). Both rest on a *heuristic* marginal-mode-dominance argument and a continuum approximation of a discrete stopping rule. Where the Route 1 depth-freedom prediction has been tested numerically—by measuring arc length of a trained neural ODE as a function of integration time, at matched, strict criticality (leading Jacobian eigenvalue driven to ≈ 0)—it was *not confirmed*: both smooth and ReLU vector fields produced *linear* arc-length growth ($R^2 \approx 1$), with ReLU no more bounded than tanh, in contradiction to (24). The interpretation is that at criticality the flow settles to approximately constant speed, giving arc length $\sim T$ independent of activation; the marginal-mode contribution does not dominate the arc length in a real finite-width, finite- T flow. An earlier “directionally consistent” observation was traced to *unmatched* criticality between the two activations and disappeared once the criticality was matched. The Route 2 scaling (20) was validated on synthetic ground truth—where a curve constructed to have a marginal $\sim \ell^{-(\alpha+1)}$ returns the exponent $-1/(\alpha+1)$ to within a few percent—but not on real networks. A subsequent sweep of trained networks (graph, sequence, and plain multilayer perceptrons) across a depth grid found the measured exponent to be *seed-unstable* in this small-network regime: where depth engages at all (a message-passing regressor, useful to ≈ 4 layers) a two-seed average yields an apparent exponent near the smooth-activation value with a convincing power-law fit, yet a three-seed average of the identical setup gives a non-monotone marginal and a nonsensical exponent. The per-layer marginal-value signal near the saturation point is dominated by seed and optimization variance rather than a clean critical power-law tail; on the compact real datasets examined, optimal depth was frequently $L^* = 1$ (depth-limited). The functional form can therefore be *fit* to any single noisy curve without that fit *confirming* the prediction. This is consistent with the criticality caveat below—real trained networks are not on the critical line—and we report it as an honest non-confirmation rather than a clean refutation.

Scope of the criticality thread. A recurring theme is that the sharpest predictions are *criticality-conditional*: they are nontrivial only on the critical line. This makes them empirically delicate, because a network trained to solve a task is generically driven off criticality, and reaching (and matching) criticality needs explicit regularization. Where criticality was reachable (the neural-ODE arc-length test), a clean verdict was obtained (refutation); where it was not (coupled multi-site selection transitions), the prediction remained untested for want of an adequate substrate.

Status of the outermost rung. The representation-dispatch level of §3 stands on a different footing from the depth-scaling corollaries. It is not a criticality-conditional scaling law but a structural claim—that interface incompatibility forecloses relaxation across contracts, forcing a discrete dispatch—together with a description-length selection rule (5) for the one ambiguous (geometric) case. That rule was tested directly: on controlled targets whose dependence was en-

gineered to be purely geometric, purely topological, or purely set-symmetric, the bake-off ranked by $\hat{R}_c + \mu_c \Omega_{\text{struct}}(c)$ selected the equivariant, plain-graph, and set contracts respectively, the physically correct choice in every case—the topological and set targets being genuine near-ties in fit that the *derived* structural code length (6) broke toward the cheaper contract. Unlike the refuted depth-freedom prediction, this outermost rung is both principled (a genuine MDL minimization, now with a derived rather than ordinal structural charge) and empirically confirmed on its intended domain; its limitation is narrowness, not fragility—it decides only the contract, while the primitive parameter α one level below it is derived separately in §9.

A Bayesian evidence reading of the contract rule. The selection $\arg \min_c [\hat{R}_c + \mu_c \Omega_{\text{struct}}(c)]$ is, once the fit term is put on a log-likelihood scale, an approximate Bayesian log-joint over contracts. Writing $\hat{L}_c$ for the mean per-datum negative log-likelihood of the fitted contract and n for the sample size, the Laplace / Bayesian-information-criterion (BIC) expansion of the evidence is $-\log p(\text{data} | c) \sim n\hat{L}_c + \frac{k_c}{2} \log n + O(1)$, and with a prior the log-joint adds $-\log p(c)$. Two facts specialize this to the contract choice. First, $\Omega_{\text{struct}}(c)$ is a description length for the contract’s *structure*, hence exactly a $-\log p(c)$ structure prior, $p(c) \propto e^{-\Omega_{\text{struct}}(c)}$. Second, the parameter-count Occam term $\frac{k_c}{2} \log n$ uses the raw weight count—the wrong complexity for singular over-parameterized networks, and moreover approximately *common* across contracts, since the bake-off trains every candidate at a shared width/depth budget—so it cancels in the posterior $\propto \exp(-[n\hat{L}_c + \Omega_{\text{struct}}(c)])$. What remains is precisely $n\hat{L}_c + \Omega_{\text{struct}}(c)$: the tie-break objective with μ_c fixed to unity (nats against nats, not a tuned exchange rate). The maximum-a-posteriori (MAP) estimate of this posterior coincides with the argmin of the priced J , so the reading does not change the winner; its value is that the fit-versus-structure tradeoff is now on a calibrated scale (the data can override the structure prior exactly when the likelihood gain warrants it) and that the argmin becomes a distribution with a reported entropy, so a near-tie is visible rather than resolved arbitrarily. The remaining gap to an exact evidence—a genuine marginal likelihood in place of the held-out $\hat{L}_c$, and the singular-model real-log-canonical-threshold (RLCT) complexity [10, 2] in place of the cancelled $\frac{k_c}{2} \log n$ —is the natural next refinement. That second half is now concrete: the package estimates the Real Log Canonical Threshold $\lambda \leq k/2$ of a fitted network by the stochastic-gradient Local Learning Coefficient of Lau et al., replacing $\frac{k}{2} \log n$ in the free energy by $\lambda \log n$. On over-parameterized networks λ is far below $k/2$ —empirically, for a widening family fitting a fixed low-rank target, λ grows sublinearly while $k/2$ grows linearly, so $\lambda/(k/2) \rightarrow 0$ —which is exactly the degeneracy that the raw parameter count, and hence classical BIC, mis-measures. The estimate is meaningful only at a converged minimum (a strongly negative λ diagnoses an unconverged w^* rather than a small complexity), which fixes both its use and its guard: the package now folds the functional charge $\lambda \log n$ into the contract objective (an opt-in refinement of the tie-break) with a clamp for slightly-negative $\hat{\lambda}$ and a validity guard that declines to price a non-converged candidate. The result completes the evidence expansion above rather than replacing it: the functional term is second-order to the structural charge on realistic converged data, and a controlled study shows it changes the selected contract only in a near-tied, anti-correlated corner (the fit-leader being the more singular contract) that natural training does not reach—so it reprices and reports effective degrees of freedom honestly, without overturning the structural verdict in the common case.

Summary. The construction is best regarded as a rigorous and pedagogically complete *map* of how variational calculus, the principle of least action, and the path-integral formalism organize neural network complexity across width and depth—exact and convex in its two-layer core, structurally faithful but only perturbatively controlled in the deep regime, and with its most quantitative depth-

scaling corollaries flagged as heuristic and, where tested at criticality, empirically overturned. The one selection variable the account had imported rather than derived, the primitive parameter α , is now brought inside the same free-energy principle (§9): it is the outer Gibbs measure $\alpha_p \propto e^{-\beta\Psi_p}$ of Eq. (13) on the primitive simplex, with DARTS its Fisher–Rao gradient flow and β the same MDL temperature as μ and λ —so the ladder from contract to weights is, save for the necessarily discrete cross-contract dispatch, one variational object throughout. The same accounting reaches one rung further out still: the *tensor shape* a flat input is given—which lattice, of which rank—is priced by the identical $R + \mu\Omega_{\text{struct}}$ with the grid-rank code length, so “which representation” joins “which contract, primitive, depth, width” as a decision of the one metaoptimality criterion rather than a preprocessing step.

10 Provenance and references

The formulations above recombine established ideas—differentiable architecture search, renormalization-group and information-bottleneck analyses of deep networks, edge-of-chaos criticality, singular learning theory, and the Kolmogorov–Arnold representation—under a single description-length action $J = R + \mu\Omega$ and a variational reading of architecture selection. Citations on first use point to the originating works; the consolidated list follows.

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